

CMP 4040 – spring 2026

ML

Phase 1 proposal

Submitted to

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[1st Choice]

Uber & Lyft Ride Price Prediction

Boston, MA — Ride-Hailing Price Forecasting

Problem Definition

Ride price prediction is a well-studied regression problem in transportation analytics, where the goal is to predict the fare of a ride given factors such as trip distance, time of day, weather conditions, and surge multipliers. The task is inherently challenging due to the dynamic nature of ride-hailing pricing, where fares fluctuate significantly based on demand and external conditions. The dataset covers real Uber and Lyft trips collected in Boston, MA, making it large enough to train robust models.

Evaluation Metrics	RMSE, R2 Score
Dataset Size	~693,000 rows, 57 features

Dataset Link: <https://www.kaggle.com/datasets/brllrb/uber-and-lyft-dataset-boston-ma>

[2nd Choice]

Football Match Outcome Prediction

Club Football Matches 2000–2025

Problem Definition

Football match outcome prediction is a well-studied classification problem in sports analytics, where the goal is to predict the result of a match — home win, draw, or away win — given historical team statistics such as recent form, goals scored and conceded, league, and season. The task is inherently challenging due to the stochastic nature of football, where upsets and draws are frequent regardless of team strength. The dataset covers 25 years of professional club football across major European leagues, making it large enough to train robust models .

Evaluation Metrics	Accuracy, F1-Score,
Dataset Size	~230,000+ matches, 48 features

Dataset Link: <https://www.kaggle.com/datasets/adamgbor/club-football-match-data-2000-2025>

[3rd Choice]

Used Car Price Prediction

Craigslist Cars & Trucks Dataset

Problem Definition

Used car price prediction is a well-studied regression problem in e-commerce analytics, where the goal is to predict the listing price of a vehicle based on attributes such as make, model, year, mileage, condition, fuel type, and location. The task is inherently challenging due to the decentralized and noisy nature of second-hand car markets, where listing quality and accuracy vary widely across sellers. The dataset covers hundreds of thousands of real vehicle listings scraped from Craigslist across the United States, making it large and diverse enough to train robust models.

Evaluation Metrics	RMSE, R2 Score
Dataset Size	~426,000 listings, 26 features

Dataset Link: <https://www.kaggle.com/datasets/austinreese/craigslist-carstrucks-data>